CHAPTER 9

THE GRAND BARGAIN

THE PROMISE OF THE STATE

At its heart, the nation-state, the central unit of the world's political order today, offers its citizens a simple and highly persuasive bargain: that not only is centralization of power in the sovereign, territorial state possible but its benefits far outweigh the risks. History suggests that a monopoly over violence—that is, entrusting the state with wide latitude to enforce laws and develop its military powers—is the surest way to enable peace and prosperity. That, moreover, a well-managed country is a key foundation of economic growth, security, and well-being. Over the last five hundred years centralizing power in a singular authority has been essential to keeping the peace, unleashing the creative talents of billions of people to work hard, seek out education, invent, trade, and, in doing so, drive progress.

Even as it grows more powerful and entangled with everyday life, the grand bargain of the nation-state, therefore, is that not only can centralized power enable peace and prosperity, but this power can be contained using a series of checks, balances, redistributions, and institutional forms. We often take for granted the delicate balance that has to be struck between extremes to maintain this. On the one hand the most dystopian excesses of centralized power must be avoided, and on the other we must accept regular intervention to maintain order.

Today, more so than at any time in history, the technologies of the coming wave threaten to unsettle this fragile equilibrium. Put simply, the

grand bargain is fracturing, and technology is a critical driver of this historic transformation.

Given that nation-states are charged with managing and regulating the impact of technology in the best interests of their populations, how prepared are they for what's to come? If the state is unable to coordinate the containment of this wave, unable to ensure it is of net benefit to its citizens, what options does that leave humanity in the medium to long term?

In the book's first two sections, we saw that a wave of powerful technologies is about to crash over us. Now it's time to consider what this means and glimpse a world after the deluge.

In this third section of the book, we grapple with the profound consequences of these technologies for the nation-state and for the liberal democratic nation-state above all. Cracks are already forming. The political order that fostered rising wealth, better living standards, growing education, science, and technology, a world tending toward peace, is now under immense strain, destabilized in part by the very forces it helped engender. The full implications are sprawling and hard to fathom, but to me they indicate a future where the challenge of containment is harder than ever, where the century's great dilemma becomes inevitable.

LESSONS FROM COPENHAGEN: POLITICS IS PERSONAL

I've always passionately believed in the power of the state to improve lives. Before my career in AI, I worked in government and the nonprofit sector. I helped start a charity telephone counseling service when I was nineteen, worked for the mayor of London, and co-founded a conflict resolution firm focused on multi-stakeholder negotiation. Working with public servants—people stretched thin and bone-tired, but forever in demand and doing heroic work for those who need it—was enough to show me what a disaster it would be if the state failed.

However, my experience with local government, UN negotiations, and nonprofits also gave me invaluable firsthand knowledge of their limitations. They are often chronically mismanaged, bloated, and slow to act. One project I facilitated in 2009 at the Copenhagen climate negotiations involved convening hundreds of NGOs and scientific experts to align their negotiating positions. The idea was to present a coherent position to 192 squabbling countries at the main summit.

Except we couldn't get consensus on anything. For starters, no one could agree on the science, or the reality of what was happening on the ground. Priorities were scattered. There was no consensus on what would be effective, affordable, or even practical. Could you raise $10 billion to turn the Amazon into a national park to absorb CO_2? How are you going to deal with the militias and bribes? Or maybe the answer was to reforest Norway, not Brazil, or was the solution to grow giant kelp farms instead? As soon as proposals were voiced, someone spoke up to poke holes in them. Every suggestion was a problem. We ended up with maximum divergence on all possible things. It was, in other words, politics as usual.

And this involved people notionally on the "same team." We hadn't even gotten to the main event and the real horse-trading. At the Copenhagen summit a morass of states all had their own competing positions. Now pile on the raw emotion. Negotiators were trying to make decisions with hundreds of people in the room arguing and shouting and breaking off into groups, all while the clock was ticking, on both the summit and the planet. I was there trying to help facilitate the process, perhaps the most complex, high-stakes multiparty negotiation in human history, but from the start it looked almost impossible. Observing this, I realized we weren't going to make sufficient progress fast enough. The timeline was too tight. The issues were too complex. Our institutions for addressing massive global problems were not fit for purpose.

I saw something similar working for the mayor of London in my early twenties. My job was to audit the impact of human rights legislation on communities in the city. I interviewed everyone from British Bangladeshis to local Jewish groups, young and old, of all creeds and backgrounds. The

experience showed how human rights law could help improve lives in a very practical way. Unlike the United States, the U.K. has no written constitution protecting people's fundamental rights. Now local groups could take problems to local authorities and point out they had legal obligations to protect the most vulnerable; they couldn't brush them under the carpet. On one level it was inspiring. It gave me hope: institutions could have a codified set of rules about justice. The system could deliver.

But of course, the reality of London politics was very different. In practice everything devolved into excuses, blame shifting, media spinning. Even when there was clear legal responsibility, departments or councils wouldn't respond, would fudge, dodge, and delay. Stasis in the face of real challenges was endemic.

Going into London's city hall, I had just turned twenty-one. It was 2005, and I was naively optimistic. I believed in local government—and the UN, for that matter; to an outsider, they seemed grand, effective institutions where we could work together to tackle the big questions. I thought, like many around that time, that globalism and liberal democracy were defaults, the welcome end state of history. Contact with reality was enough to show the gulf between hopeless ideals and the facts on the ground.

Around that time, I also started to pay attention to something else then taking shape. Facebook was growing at unprecedented speed. Somehow, even as everything from local government to the UN seemed to operate at a glacial pace, this small start-up had grown to more than 100 million monthly users in just a few years. That single fact changed the course of my life. It was very clear to me that some organizations were still capable of highly effective action at scale and they were operating in new spaces, like online platforms.

The idea that technology alone can solve social and political problems is a dangerous delusion. But the idea that they can be solved without technology is also wrongheaded. Seeing the frustrations of public servants up close made me want to find other effective ways to get things done at scale, working not against but in concert with the state to make more productive, fairer, kinder societies.

Technological breakthroughs will help us meet the challenges hinted at in the last section: grow food amid unsustainable temperatures; detect floods, earthquakes, and fires ahead of time; and increase the standard of living for everyone. At a time of spiraling costs and deteriorating services, I see AI and synthetic biology as critical levers to help accelerate progress. They will make health care both higher-quality and more affordable. They will help us invent tools to bring about the transition to renewable energy and combat climate change at a time when politics has stalled, and support teachers, helping to increase the effectiveness of underfunded education systems. This is the real potential of the next wave.

So I embarked on a career in technology, believing that a new generation of tools could amplify our ability to act at scale, operating far more rapidly than traditional policies. Putting them to work to "invent the future" seemed like the best way to spend the most productive years of my life.

I invoke my idealistic streak to put the following chapters into context, to make clear that I regard the often dismal picture painted as a titanic failure of technology and a failure of people like me who build it.

While technology is still the single most powerful avenue for addressing the challenges of the twenty-first century, we cannot ignore downsides. While acknowledging the many benefits, we also must overcome pessimism aversion and take a cold, hard look at what new risks might arise from omni-use technologies. Over time the nature of those risks—and the size of the stakes—have only become clearer. Technology is not just a tool to support the bargain we've made in the nation-state; it is also a genuine threat to it.

An influential minority in the tech industry not only believes that new technologies pose a threat to our ordered world of nation-states; this group actively welcomes its demise. These critics believe that the state is mostly in the way. They argue it's best jettisoned, already so troubled it is beyond rescue. I fundamentally disagree; such an outcome would be a disaster.

I'm British, born and raised in London, but one side of my family is Syrian. My family has been caught up in the terrible war suffered by that

country in recent years. I know well what it looks like when states fail, and to put it crudely, it's unimaginably bad. Horrific. And anyone who thinks what happened in Syria could never happen "here" is kidding themselves; people are people wherever they are. Our system of nation-states isn't perfect, far from it. Nonetheless, we must do everything to bolster and protect it. This book, in part, is my attempt to rally to its defense.

Nothing else—no other silver bullet—will arrive in time to save us, to absorb the destabilizing force of the wave. There simply isn't another option in the medium term.

Even in best-case scenarios the coming wave will be an immense shock to the systems governing societies. Before we explore the perils of the wave, it's worth asking about the broad health of nation-states. Are they in any shape to meet the challenges ahead?

FRAGILE STATES

Global living conditions are objectively better today than at any time in the past. We take running water and plentiful food supplies for granted. Most people enjoy warmth and shelter all year round. Literacy rates, life expectancy, and gender equality sit at all-time highs. The sum of thousands of years of human scholarship and inquiry is available at the touch of a button. For most people in developed countries, life is marked by an ease and abundance that would have seemed unbelievable in bygone eras. And yet, under the surface, there's a nagging feeling that something isn't quite right.

Western societies in particular are mired in a deep-seated anxiety; they are "nervous states," impulsive and fractious. This persistent unease is partly the result of previous shocks—multiple financial crises, the pandemic, violence (everything from 9/11 to the Ukraine war)—and partly the effect of long-term and growing pressures like declining public trust, rising inequality, and a warming climate. Going into the coming wave, many nations are beset by a slew of major challenges battering their

effectiveness, making them weaker, more divided, and more prone to slow and faulty decision-making. The coming wave will land in a combustible, incompetent, overwrought environment. This makes the challenge of containment—of controlling and directing technologies so they are of net benefit to humanity—even more daunting.

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DEMOCRACIES ARE BUILT ON trust. People need to trust that government officials, militaries, and other elites will not abuse their dominant positions. Everyone relies on the trust that taxes will be paid, rules honored, the interests of the whole put ahead of individuals. Without trust, from the ballot box to the tax return, from the local council to the judiciary, societies are in trouble.

Trust in government, particularly in America, has collapsed. Postwar presidential administrations like those of Eisenhower and Johnson were trusted to do "what is right" by more than 70 percent of Americans, according to a Pew survey. For recent presidents such as Obama, Trump, and Biden, this measure of confidence has cratered, all falling below 20 percent. Quite remarkably, a 2018 study of democracy in America found that as many as one in five believe "army rule" is a good idea! No less than 85 percent of Americans feel the country is "heading in the wrong direction." Distrust extends to nongovernment institutions, with growing levels of distrust in the media, the scientific establishment, and the idea of expertise in general.

The problem is not limited to the United States. Another Pew survey found that across twenty-seven countries, a majority were dissatisfied with their democracies. A Democracy Perception Index poll found that across fifty nations two-thirds of respondents felt the government "rarely" or "never" acted in the public interest. That so many people profoundly feel society is failing is itself a problem: Distrust breeds negativity and apathy. People decline to vote.

Since 2010, more countries have slid backward on measures of democracy than have progressed, a process that appears to be accelerating. Rising nationalism and authoritarianism seem endemic, from Poland and China to Russia, Hungary, the Philippines, and Turkey. Populist movements range from the bizarre, like QAnon, to the directionless (the *gilets jaunes* in France), but from Bolsonaro in Brazil to Brexit in the U.K. their prominence on the world stage has been impossible to miss.

Behind the new authoritarian impulse and political instability lies a growing pool of social resentment. A key catalyst of instability and social resentment, inequality has surged across Western nations in recent decades, and nowhere more so than in the United States. Between 1980 and 2021 the share of national income earned by the top 1 percent has almost doubled and now sits just under 50 percent. Wealth is ever more concentrated in a tiny clique. Government policy, a shrinking working-age population, stalling educational levels, and decelerating long-term growth have all contributed to decisively more unequal societies. Forty million people in the United States live in poverty, and more than five million live in "Third World conditions"—all within the world's richest economy.

These are especially worrying trends when you consider persistent relationships between social immobility, widening inequality, and political violence. Across data from more than one hundred countries, evidence suggests that the lower a country's social mobility, the more it experiences upheavals like riots, strikes, assassinations, revolutionary campaigns, and civil wars. When people feel stuck, that others are unfairly hogging the rewards, they get angry.

Not so long ago, the world was meant to be "flat"—a frictionless terrain of easy trade and rising prosperity. In fact, as the twenty-first century wears on, supply chain crunches and financial shocks remain indelible features of the economy. Those countries leaning into nationalism are, in part, experiencing a turning away from the bright twentieth-century promise that greater interconnectedness would accelerate the spread of wealth and democracy.

Onshoring, national security, resilient supply chains, self-sufficiency—today's language of trade is once again the language of borders, barriers, and tariffs. At the same time, food, energy, raw materials, and goods of all kinds have become more expensive. Essentially the entire postwar security and economic order is facing unprecedented strain.

Global challenges are reaching a critical threshold. Rampant inflation. Energy shortages. Stagnant incomes. A breakdown of trust. Waves of populism. None of the old visions from either left or right seem to offer convincing answers, yet better options seem in short supply. It would take a brave, or possibly delusional, person to argue that all is well, that there are not serious forces of populism, anger, and dysfunction raging across societies—all despite the highest living standards the world has ever known.

This makes containment far more complicated. Forming national and international consensus and establishing new norms around fast-moving technologies are already steep challenges. How can we hope to do this when our baseline mode seems to be instability?

TECHNOLOGY IS POLITICAL: THE WAVE'S CHALLENGE TO STATES

Every previous wave of technology has had profound political implications. We should expect the same in the future. The last wave—the arrival of mainframes, desktop PCs and desktop software, the internet, and the smartphone—delivered immense benefits to society. It laid down the new tools for the modern economy, bolstering growth, transforming access to knowledge, to entertainment, and to one another. Amid the present hand-wringing about the negative effects of social media, it's easy to overlook these myriad positives. Yet over the last decade a growing consensus suggests these technologies did something else as well: creating the conditions to feed and amplify this underlying political polarization and institutional fragility.

It's hardly news that social media platforms can trigger gut emotional responses, the jolts of adrenaline so effectively delivered by perceived threats. Social media thrives on heightened emotions and, quite often, outrage. A meta-analysis published in the journal *Nature* reviewed the results of nearly five hundred studies, concluding there is a clear correlation between growing use of digital media and rising distrust in politics, populist movements, hate, and polarization. Correlation may not be causation, but this systematic review throws up "clear evidence of serious threats to democracy" coming from new technologies.

Technology has already eroded the stable, sovereign borders of nation-states, creating or supporting innately global flows of people, information, ideas, know-how, commodities, finished goods, capital, and wealth. It is, as we have seen, a significant component of geopolitical strategy. It touches on almost every aspect of people's lives. Even before the coming wave hits, technology is a driver on the world stage, a major factor in the deteriorating health of nation-states around the world. Too fast in its development, too global, too protean and enticing for any simple model of containment, strategically critical, relied upon by billions, modern technology itself is a prime actor, a monumental force nation-states struggle to manage. AI, synthetic biology, and the rest are being introduced to dysfunctional societies *already* rocked back and forth on technological waves of immense power. This is not a world ready for the coming wave. This is a world buckling under the existing strain.

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I'VE OFTEN HEARD IT said that technology is "value neutral" and that its politics arise from its use. This is so reductive and simplistic that it's almost meaningless. Technology didn't straightforwardly "cause" or create the modern state (or indeed any political structure). But the potential it unleashes is not neutral in that story.

As the historian of technology Langdon Winner puts it, "Technology in its various manifestations is a significant part of the human world. Its

structures, processes, and alterations enter into and become part of the structures, processes, and alterations of human consciousness, society, and politics." In other words, technology is political.

This fact is radically under-recognized not only by our leaders but even by those building the technology itself. At times this subtle but omnipresent politicization is nearly invisible. It shouldn't be. Social media is just the most recent reminder that technology and political organization cannot be divorced. States and technologies are intimately tied together. This has important ramifications for what's coming.

While technology doesn't simplistically push people in a predetermined direction, it's not naive techno-determinism to recognize its tendency to afford certain capabilities or see how it prompts some outcomes over others. In this, technology is one of the key determinants of history, but never alone and never in a mechanistic, inherently predictable way. It doesn't superficially cause given behaviors or outcomes, but what it produces does guide or circumscribe possibilities.

War, peace, commerce, political order, culture—these have always been fundamentally interlinked, and interlinked moreover with technology. Technologies are ideas, manifested in products and services that have profound and lasting consequences for people, social structures, the environment, and everything in between.

Modern technology and the state evolved symbiotically, in constant dialogue. Think of how technology facilitated the state's core working parts, helping construct the edifice of national identity and administration. Writing was invented as an administrative and accounting tool to keep track of debts, inheritances, laws, taxes, contracts, and records of ownership. The clock produced set times, first in limited spaces like monasteries but then in mechanical form across late medieval mercantile cities and eventually across nations, creating common, and ever larger, social units. The printing press helped standardize national languages from a chaos of dialects and thus helped produce a national "imagined community," the unitary people behind a nation-state. Supplanting more fluid oral traditions, the printed word fixed geography, knowledge, and history in place, promulgating set

legal codes and ideologies. Radio and TV turbocharged this process, creating moments of national and even international commonality experienced simultaneously, like FDR's fireside chats or the World Cup.

Weapons, too, are technologies central to the power wielded by nation-states. Indeed, theorists of the state often suggest that war itself was foundational to its creation (in the words of the political scientist Charles Tilly, "War made the state and the state made war"), just as conflict has always been a spur to new technologies—from chariots and metal armor to radar and the advanced chips that guide precision munitions. Introduced to Europe in the thirteenth century, gunpowder broke the old pattern of defensive medieval castles. Fortified settlements were now sitting ducks for bombardment. By the Hundred Years' War between Britain and France, offensive capabilities gave the advantage to those who could afford to buy, build, maintain, move, and deploy capital-intensive cannons. Over the years, the state concentrated ever-increasing lethal power in its own hands, claiming a monopoly on the legitimate use of force.

Put simply, technology and political order are intimately connected. The introduction of new technologies has major political consequences. Just as the cannon and the printing press upended society, so we should expect the same from technologies like AI, robotics, and synthetic biology.

Pause for a moment and imagine a world where robots with the dexterity of human beings that can be "programmed" in plain English are available at the price of a microwave. Can you begin to think of all the uses to which such a valuable technology will be put? Or how widely such tools will be adopted? Who or rather what will be looking after your elderly mother at a care home? How will you order food at a restaurant, and who will bring it to your table? What does law enforcement look like in a hostage situation? Who will staff orchards at harvest time? How will military and paramilitary planners react when no humans need be sent into combat? What will the sports field be like when kids are training at football? What will your window cleaner look like? Who owns all this hardware and IP, who controls it, what safeguards are in place for if—when—it goes wrong?

Imagine all this, and it implies a very different political economy from today's.

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THE MODERN, LIBERAL DEMOCRATIC industrialized nation-state has been the dominant global force since the early twentieth century, the clear "victor" of last century's great political clash. It came with defining functions now taken for granted. The provision of security. Great concentrations of legitimate power at the center, capable of utterly dominating within their jurisdictions, but also sensible checks and balances on, and separations between, all forms of power. Adequate welfare via redistribution and sound economic management. Stable frameworks of technological innovation and regulation, alongside a whole socioeconomic-legal architecture of globalization.

In the next few chapters we will see how the coming wave places all of this under great threat.

What emerges will, I think, tend in two directions with a spectrum of outcomes in between. On one trajectory, some liberal democratic states will continue to be eroded from within, becoming a kind of zombie government. Trappings of liberal democracy and the traditional nation-state remain, but functionally they are hollowed out, the core services increasingly threadbare, the polity unstable and fractious. Lurching on in the absence of anything else, they become ever more degraded and dysfunctional. On another, unthinking adoption of some aspects of the coming wave opens pathways to domineering state control, creating supercharged Leviathans whose power goes beyond even history's most extreme totalitarian governments. Authoritarian regimes may also tend toward zombie status, but equally they may double down, get boosted, become fully fledged techno-dictatorships. On either path, the delicate balance holding states together is tipped into chaos.

Both failing states and authoritarian regimes are disastrous outcomes, not just on their own terms, but also for governing technology; neither

flailing bureaucracies, populist opportunists, nor all-powerful dictators are people you'd want to be fundamentally responsible for controlling powerful new technologies. Neither direction can or will contain the coming wave.

On either side, then, lies danger, given that managing the coming wave requires confident, agile, coherent states, accountable to the people, filled with expertise, balancing interests and incentives, capable of reacting fast and decisively with legislative action and, crucially, close international coordination. Leaders will need to take bold actions without precedent, trading off short-term gain for long-term benefit. Responding effectively to one of the most far-reaching and transformative events in history will require mature, stable, and most of all trusted governments to perform at their best. States that work really, really well. That is what it will take to ensure that the coming wave delivers the great benefits it promises. It's an incredibly tall order.

Cheap, omnipresent robots like those sketched above are, alongside a host of other transformative technologies we saw in part 2, utterly inevitable over a twenty-year horizon, and possibly much sooner. In this context we should expect profound changes to the economy, the nation-state, and everything that goes with them. The grand bargain is already in trouble. As the deluge begins, a series of new stressors will shake its foundations.